



Thoracic Surgery Data

Donated on 11/12/2013

The data is dedicated to classification problem related to the post-operative life expectancy in the lung cancer patients: class 1 - death within one year after surgery, class 2 - survival.

Dataset Characteristics

Multivariate

Subject Area

Health and Medicine

Associated Tasks

Classification

Feature Type

Integer, Real

Instances

470

Features

16

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DOI

10.24432/C5Z60N

License

Dataset Information

What do the instances in this dataset represent?

Individual patients

Additional Information

The data was collected retrospectively at Wroclaw Thoracic Surgery Centre for patients who underwent major lung resections for primary lung cancer in the years 2007 to 2011. The Centre is associated with the Department of Thoracic Surgery of the Medical University of Wroclaw and Lower-Silesian Centre fo...

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Has Missing Values?
No

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Introductory Paper



[Boosted SVM for extracting rules from imbalanced data in application to prediction of the post-operative life expectancy in the lung cancer patients](#)
By Maciej Zięba, Jakub M. Tomczak, M. Lubicz, J. Swiatek. 2014
Published in Applied Soft Computing

This allows for the sharing and adaptation of the datasets for any purpose, provided that the appropriate credit is given.

Variables Table



Variable Name	Role	Type	Demographic	Description	Units	Missing Va
DGN	Feature	Categorical		Diagnosis - specific combination of ICD-10 codes for primary and secondary as well multiple tumours if any		no
PRE4	Feature	Continuous		Forced vital capacity - FVC (numeric)		no
PRE5	Feature	Continuous		Volume that has been exhaled at the end of the first second of forced expiration - FEV1 (numeric)		no

Variable Name	Role	Type	Demographic	Description	Units	Missing Va
PRE6	Feature	Categorical		Performance status - Zubrod scale (PRZ2,PRZ1,PRZ0)		no
PRE7	Feature	Binary		Pain before surgery (T,F)		no
PRE8	Feature	Binary		Haemoptysis before surgery (T,F)		no
PRE9	Feature	Binary		Dyspnoea before surgery (T,F)		no
PRE10	Feature	Binary		Cough before surgery (T,F)		no
PRE11	Feature	Binary		Weakness before surgery (T,F)		no
PRE14	Feature	Categorical		T in clinical TNM - size of the original tumour, from OC11 (smallest) to OC14 (largest) (OC11,OC14,OC12,OC13)		no
PRE17	Feature	Binary		Type 2 DM - diabetes mellitus (T,F)		no
PRE19	Feature	Binary		MI up to 6 months (T,F)		no

Variable Name	Role	Type	Demographic	Description	Units	Missing Va
PRE25	Feature	Binary		PAD - peripheral arterial diseases (T,F)		no
PRE30	Feature	Binary		Smoking (T,F)		no
PRE32	Feature	Binary		Asthma (T,F)		no
AGE	Feature	Integer	Age	Age at surgery (numeric)		no
Risk1Yr	Target	Binary		1 year survival period - (T)rue value if died (T,F)		no

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Additional Variable Information



- 1. DGN: Diagnosis - specific combination of ICD-10 codes for primary and secondary as well multiple tumours if any (DGN3,DGN2,DGN4,DGN6,DGN5,DGN8,DGN1)
- 2. PRE4: Forced vital capacity - FVC (numeric)
- 3. PRE5: Volume that has been exhaled at the end of the first second of forced expiration - FEV1 (numeric)
- 4. PRE6: Performance status - Zubrod scale (PRZ2,PRZ1,PRZ0)
- 5. PRE7: Pain before surgery (T,F)
- 6. PRE8: Haemoptysis before surgery (T,F)
- 7. PRE9: Dyspnoea before surgery (T,F)
- 8. PRE10: Cough before surgery (T,F)
- 9. PRE11: Weakness before surgery (T,F)

- 10. PRE14: T in clinical TNM - size of the original tumour, from OC11 (smallest) to OC14 (largest) (OC11,OC14,OC12,OC13)
- 11. PRE17: Type 2 DM - diabetes mellitus (T,F)
- 12. PRE19: MI up to 6 months (T,F)
- 13. PRE25: PAD - peripheral arterial diseases (T,F)
- 14. PRE30: Smoking (T,F)
- 15. PRE32: Asthma (T,F)
- 16. AGE: Age at surgery (numeric)
- 17. Risk1Y: 1 year survival period - (T)rue value if died (T,F)

Class Distribution: the class value (Risk1Y) is binary valued.

Risk1Y Value: Number of Instances:

T	70
N	400

Summary Statistics:

Binary Attributes Distribution:

PRE7 Value: Number of Instances:

T	31
N	439

PRE8 Value: Number of Instances:

T	68
N	402

PRE9 Value: Number of Instances:

T	31
N	439

PRE10 Value: Number of Instances:

T	323
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N 147

PRE11 Value: Number of Instances:

T 78

N 392

PRE17 Value: Number of Instances:

T 35

N 435

PRE19 Value: Number of Instances:

T 2

N 468

PRE25 Value: Number of Instances:

T 8

N 462

PRE30 Value: Number of Instances:

T 386

N 84

PRE32 Value: Number of Instances:

T 368

N 2

Nominal Attributes Distribution:

DGN Value: Number of Instances:

DGN3 349

DGN2 52

DGN4 47

DGN6 4

DGN5 15

DGN8 2

DGN1 1

PRE6 Value: Number of Instances:

PRZ2	27
PRZ1	313
PRZ0	130

PRE14 Value: Number of Instances:

OC11	177
OC14	17
OC12	257
OC13	19

Numeric Attributes Statistics:

	Min	Max	Mean	SD
PRE4:	1.4	6.3	3.3	0.9
PRE5:	0.96	86.3	4.6	11.8
AGE:	21	87	52.5	8.7

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Dataset Files ^

File	Size
ThoracicSurgery.arff	23.7 KB

Reviews ^

There are no reviews for this dataset yet.

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